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Project Presentation of Research Project COOP-II (22CS421)

On

A Comparative Study of Image Compression Techniques Across Traditional and Modern Formats

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Introduction

The explosive growth of digital content has created unprecedented demand for efficient image storage, transmission, and rendering across modern computing systems.

Background

- Millions of images generated daily across social media, e-commerce & cloud platforms.
- Images dominate web-based services and real-time communication apps.
- Reducing image size saves bandwidth, storage cost & boosts user experience.

Need for Better Formats

- JPEG / PNG: limited compression efficiency & visible artifacts at low bit-rates.
- Modern formats (WebP, AVIF, JPEG XL) use advanced video-codec techniques.
- Perceptual modeling & predictive coding deliver smaller files at higher quality.

Goal: *Compare 5 image formats under fair, reproducible conditions — objective + perceptual metrics.*

Objectives of the Research

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Five clearly defined goals drive this comparative research:

01

Theoretical Comparison

Detailed survey of compression algorithms across JPEG, PNG, WebP, AVIF & JPEG XL.

02

Standardized Testing

Experimental evaluation using uniform datasets, parameters & encoding tools.

03

Dual Metric Evaluation

Objective metrics (PSNR, SSIM) combined with perceptual quality (Butteraugli).

04

Image-wise Variability

Analyze how compression performance varies per image — a key novelty.

05

Application Mapping

Recommend the best format for web, archival, and high-quality imaging use cases.

Image Compression Formats

JPEG *Lossy • 1992*	DCT (8×8 blocks) + Huffman coding. Universally compatible but produces blocking artifacts at high compression.
PNG *Lossless*	DEFLATE compression with predictive filtering. Perfect fidelity but very large file sizes.
WebP *Lossy / Lossless*	Google's VP8-based format. Intra prediction + arithmetic coding. Better than JPEG but variable.
AVIF *Lossy / Lossless*	AV1 video codec base. Advanced prediction + context-adaptive coding. Best compression efficiency.
JPEG XL *Lossy / Lossless*	Hybrid transforms + adaptive prediction + ANS entropy coding. Highest perceptual quality.

Methodology & Pipeline

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Dataset Preparation

- **Diverse content:** high-frequency textures, smooth gradients, sharp edges, complex colors.
- **Normalization:** resolution standardized to 1024 × 1024, uniform color space, lossless source.

Compression Pipeline

1	**2**	**3**	**4**
Preprocess	**Compress**	**Decode**	**Evaluate**
Resize, validate, standardize	Encode into all 5 formats	Standardize to PNG	PSNR, SSIM, Butteraugli, Size

Experimental Setup & Metrics

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Evaluation Metrics

PSNR

Peak Signal-to-Noise Ratio

Pixel-level reconstruction accuracy.

SSIM

Structural Similarity Index

Luminance, contrast & structural similarity.

Butteraugli

Perceptual Metric

Models human visual perception of differences.

File Size

Compression Cost

Storage & transmission efficiency.

Encoding Tools & Parameters

JPEG
FFmpeg • Q = 75

WebP
cwebp • Q = 75

AVIF
avifenc • CQ = 30

JPEG XL
cjxl • CQ = 30

Results and Comparative Analysis

Format	File Size	PSNR (dB)	SSIM	Butteraugli
AVIF	~100 KB	36.44	0.919	Low
JPEG XL	~134 KB	37.04	0.9196	2.46 (Best)
WebP	~113 KB	36.00	0.917	3.38
JPEG	~156 KB	36.25	0.917	2.85

AVIF

35–55% smaller files than JPEG at comparable quality

JPEG XL

Highest fidelity & most consistent across image types

JPEG

Stable but outdated — largest files, no clear advantage

Conclusion and Applications

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Key Findings

- **AVIF —** best compression efficiency.
- **JPEG XL —** highest perceptual quality & consistency.
- **WebP —** balanced but high variability.
- **JPEG —** stable but declining in relevance.

Applications

- **Web & Mobile:** AVIF, WebP for fast loads.
- **Archival:** PNG, JPEG XL for fidelity.
- **Photography:** JPEG XL for fine detail.
- **Legacy:** JPEG for compatibility.

Future Scope

- **Content-adaptive** compression tuning.
- **Real-time** encoding performance studies.

Format selection should be guided by the application context, not by averaged metrics alone.

Thank You